Distributed Systems
Homework task #1

Variant 2

Multithreaded Text Processing with Log Collection (Python)

Description

The aim of this homework project is to design and implement a **multithreaded text processing system in Python** that:

1. Processes a large text file in parallel.
2. Computes global word statistics.
3. Simultaneously generates and collects system logs.

The project must demonstrate concurrency, synchronization, and aggregation of distributed results.

Steps – Part 1: Parallel Large Text Processing

1. Choose any text file with amount of lines non-less than 10 000.
2. Split the text into chunks.
3. Process chunks in parallel using threads.
4. Aggregate final results correctly.

The system must compute:

1. Total word count.
2. Unique word count.
3. Top 20 most frequent words.

After all threads finish:

1. Merge results into global statistics.
2. Output:

- Total word count
- Total unique word count
- Top 20 most frequent words

Steps – Part 2: Log Production and Collection

While processing the text, the system must also simulate internal system logging.

Each worker thread must generate logs such as:

- `"Worker 2 started processing chunk"`
- `"Worker 2 finished chunk in 0.45 seconds"`
- `"Worker 3 processed 120000 words"`
- `"Worker 1 encountered empty line"`

Each log entry must include:

- Timestamp
- Worker ID
- Log level (INFO, WARNING, ERROR)
- Message

Logs must be sent to a shared thread-safe queue.

Log Collector

A separate **log collector thread** must:

- Continuously read logs from the queue
- Count:
  - Total log messages
  - Logs per level (INFO/WARNING/ERROR)
- Optionally store logs to file

At the end, print log statistics.

Technical Constraints

- Use Python standard library only
- Use `threading`
- Use `queue.Queue` for log communication
- Ensure proper synchronization
- Avoid race conditions

General Requirements

1. Use threading or ThreadPoolExecutor.
2. Each thread processes one chunk.
3. Measure execution time (single-thread vs. multi-thread)

Final Program Output Example

```
TEXT STATISTICS
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Total words: 5,200,000
Unique words: 145,320
Top 20 words:
1. the
2. and
3. ...

LOG STATISTICS
--------------
Total logs generated: 120
INFO: 100
WARNING: 18
ERROR: 2

Execution time: 3.8 seconds
Throughput: 1,368,421 words/sec
```

References

1. Python threading Module
2. queue — A synchronized queue class
3. Multithreading in Python: A Comprehensive Guide (Beginner to Advanced)